

Li Shunyang

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Skills

- **Languages:** C++ (C++17/20, templates, concurrency), Python, Bash
- **Performance & Quantization:** CUDA, Triton, TensorRT, ONNX, FP8/NVFP4/W4A16 (ModelOpt, AutoRound), low-latency optimization
- **ML & Inference:** PyTorch, vLLM / vLLM-Omni, diffusion & omni-modality serving

Open-Source Contributions

vLLM-Omni Committer

- Own the **quantization**, **config**, **deploy**, and **metrics** modules of **vLLM-Omni** (4.8k+ stars), a high-throughput inference framework for omni-modality LLMs; landed **27 merged PRs** spanning core infrastructure, model support, profiling, and quantization.
- Designed and merged a **unified quantization framework** (#1764) that lets omni and diffusion models run in lower precision.
- Authored the **Continuous Quantization Support** RFC (#1854) tracking quantization coverage across image, video, audio, and omni models; driving **ModelOpt FP8** quantization for video diffusion models—Wan2.2, HunyuanVideo-1.5 (#2924, #2920)—and **AutoRound W4A16** support.
- Optimized diffusion inference on **NVIDIA Blackwell B300 GPUs**: **FlashInfer/cuDNN** attention backends with auto-routing (#3079), **SageAttention3** FP4 attention, and **ModelOpt NVFP4** quantization (#3305).
- Led a multi-PR **configuration-system refactor** (#1115, #2383, #2915); added **HunyuanVideo-1.5** video-generation support (#1516) and a **Qwen3-TTS** demo for online serving (#1231).
- Built **profiling and progress-tracking tools** for the inference pipeline (#709, #552) with developer guides (#738, #570), helping users measure and tune model performance.

vLLM Contributor

- Drove early adoption of **TurboQuant** (ICLR 2026, Google Research) in **core vLLM** (#38280), adding it as a new `--kv-cache-dtype` backend (Hadamard rotation + Lloyd–Max scalar quantization + outlier-aware channel allocation with fused Triton encode/decode kernels).

Industry Experience

Backend Engineer Intern, Recommendation Architecture ByteDance / TikTok, Singapore Mar 2026 – Present

- Built the end-to-end **data pipeline** for a **unified LLM-generation framework** powering **q2q (query-to-query)** and **v2q (video-to-query)** generation tasks within TikTok's recommendation architecture.
- Built scalable offline data pipelines (ingestion, cleaning, and feature processing) on the team's large-scale infrastructure to produce training data for the framework.

LLM Engineer Intern Thales DIS, Singapore May 2025 – Dec 2025

- Scaled **vLLM** inference for **Qwen3-Coder-30B / Qwen3-30B-A3B-Instruct** on **2×H100 80GB GPUs**, cutting total inference time for 14k samples from **26 hours to 15 minutes** ($\approx 104\times$ **speedup**).
- Fine-tuned models with **QLoRA** for code generation and completion, improving accuracy on internal JavaCard and cryptography benchmarks; co-built a dataset generator producing 10k+ training samples.

Computer Vision Lead (Extracurricular) Calibur Robotics, NUS Robomasters, Singapore Aug 2024 – Oct 2025

- Achieved **end-to-end perception latency of 2.2 ms/frame** on **NVIDIA Jetson AGX Thor** by designing a real-time pipeline spanning camera capture, **YOLOv11-Pose** detection, pose estimation, target tracking, and command output.
- Optimized the model with **ONNX** and **TensorRT** plus custom GPU kernels, reaching **1.7 ms** inference latency on Jetson AGX Thor.

Education

National University of Singapore Aug 2023 – May 2027

- *Bachelor of Engineering in Computer Engineering (Honours), Minor in Mathematics*
- Cumulative GPA: 4.58/5.0